

Karib Shams

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93 South Bashabo, Dhaka-1214, Bangladesh

EDUCATION

- **East West University** Dhaka, Bangladesh
Master of Science in Computer Science (Data Science); CGPA: 3.91/4.00 Jan. 2025 – Dec. 2025
- **East West University** Dhaka, Bangladesh
Bachelor of Science in Computer Science; CGPA: 3.58/4.00 Jan. 2020 – Jul. 2024
- **National Ideal College** Dhaka, Bangladesh
Higher Secondary Certificate (HSC); GPA: 4.67/5.00 2017 – 2019
- **Motijheel Model School And College** Dhaka, Bangladesh
Secondary School Certificate (SSC); GPA: 5.00/5.00 2016 – 2017

EXPERIENCE

- **Join Venture Ai (JVai)** Dhaka, Bangladesh
Senior Executive Data Scientist & Team Leader Jun. 2025 - Oct. 2026
 - **AI System Development:** Developed 13+ production AI systems, including RAG-based NLP chatbots, cutting client response time by 40%.
 - **Team Leadership:** Led a 45+ member AI team across R&D sprints, technical planning, and delivery, completing client-facing features on schedule.
 - **Workflow Automation:** Automated n8n workflows with webhook triggers and OpenAI/email APIs, cutting manual processing by 10 hrs/week.
 - **Sales Enablement:** Created AI-enabled lead generation tools with the sales team, improving pipeline efficiency and prospect qualification.
- **East West University** Dhaka, Bangladesh
Graduate Teaching Assistant (GTA) Oct. 2024 – Dec. 2025
 - **Instruction:** Taught lectures and lab sessions for Statistics, AI, and Machine Learning courses to a cohort of 70+ graduate students per semester.
 - **Mentorship:** Guided 90+ students in advanced ML projects and data analysis assignments, contributing to improved student performance outcomes.
- **East West University** Dhaka, Bangladesh
Research Assistant Oct. 2024 – Dec. 2025
 - **Academic Research:** Co-authored 17 peer-reviewed publications in Data Science, Computer Vision, and NLP, with 1 Best Paper Award at an international conference.
 - **Dataset Contribution:** Supported development of the TFP-BD dataset for traffic flow and pedestrian analysis, published in Data in Brief (Vol. 59, 2025).

PROJECTS

- **AI-Powered Invoice Voucher Processing System – Freelance (Fiverr):** Architected a GPT-4o Vision pipeline that extracts invoice data, classifies GL accounts and profit centres, and generates payment vouchers with confidence-based review. [Live Demo]
- **BhromonGhuri – Full-Stack Travel & Tour Booking Platform:** Built a full-scale tourism platform (Django 5, HTMX, Alpine.js) with zero-reload filtering, instant pricing, and bKash/Nagad payment verification. [Live on October]
- **HealthRide – AI-driven NEMT Platform:** Delivered a 24/7 GPT-4o voice receptionist and AI dispatch engine with real-time driver matching across 6 event triggers.
- **Eat at Home – AI Meal Cost Estimator:** Designed a GPT-4o meal-cost estimation model with store-tier blending and ZIP-code regional pricing multipliers. [View Project]
- **EmoThrive – AI Therapeutic Assistant:** Launched a voice-enabled LLM therapy assistant with PDF-backed knowledge retrieval, integrating speech recognition and a RAG workflow. [Try Demo]
- **RAG-Based AI System:** Deployed a Retrieval-Augmented Generation pipeline using FAISS embeddings and OpenAI LLMs, delivering context-aware responses across 500+ document chunks.

- **n8n AI Video Generation Automation:** Orchestrated end-to-end AI video creation using n8n, cutting manual production effort by ~80% through API-chained workflows.
- **Hebrew/Yiddish Handwriting Transcription System:** Engineered an OCR pipeline for cursive handwriting using GPT-4o and Claude Vision APIs with CER-based accuracy evaluation. [Live Demo]
- **OCR Text Extraction System:** Assembled a scalable OCR engine supporting multi-format document ingestion and structured text output. [GitHub]
- **Nory0929 – Video-Based Auto Marketplace:** Produced a multimodal video generation agent using Kling 2.5 and OpenAI that transforms 5+ vehicle images and an optional prompt into 25-second, 720p marketing videos.
- **Vehicle Detection and Traffic Flow Prediction:** Trained deep learning models on the TFP-BD dataset for vehicle detection and traffic flow prediction on Bangladeshi urban roads.

PUBLICATIONS

- **CodeMixEcom-Emotion:** A. Das, K. Shams et al., “CodeMixEcom-Emotion: A Large-Scale Bangla–English Review Corpus and Transformer-Based Benchmark for Fine-Grained Emotion Detection.” AII 2025, Springer-Nature CCIS. [Best Paper Award – Washington, D.C., USA]
- **Kidney CT Scan Classification:** A. Das, K. Shams et al., “Towards Annotation-Efficient Kidney CT Scan Classification: Supervised and Semi-Supervised Swin Transformer Frameworks.” IEEE SPICSCON 2025. (Accepted)
- **Small Cell Lung Cancer:** K. Shams et al., “Histopathology Images-Based Deep Learning Prediction of Prognosis and Therapeutic Response in Small Cell Lung Cancer.” ICDMIS 2024, Springer.
- **TFP-BD Dataset:** M.M. Islam, A. Das, K. Shams et al., “TFP-BD: An Image Dataset for Traffic Flow and Pedestrian Movement Analysis on Bangladeshi Urban Roads.” Data in Brief, Vol. 59, p. 111398, (2025).
- **Tuberculosis Diagnosis:** K. Shams et al., “Tuberculosis Diagnosis from Chest X-Ray Image Using Deep Learning Techniques.” IEEE ICAECT 2025, pp. 1–6.
- **Oyster Mushroom Monitoring:** R. Ahmed, K. Shams et al., “Real-Time Monitoring of Oyster Mushroom Cultivation Using CCTV and Attention-Enhanced ShuffleNet-Based Explainable AI.” Smart Agricultural Technology, Vol. 12, p. 101571, (2025).

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, C, C++, Java, HTML, CSS **Frameworks:** Django, FastAPI, TensorFlow, PyTorch, Scikit-learn, Streamlit
- **AI / ML:** Deep Learning, NLP, Computer Vision, OCR, RAG Pipelines, LLMs, Generative AI, Explainable AI (SHAP), Knowledge Graphs
- **Automation & MLOps:** n8n, LLM Chaining, Webhook Systems, OpenAI API, FAISS, Prompt Engineering
- **Tools:** Jupyter Notebook, Google Colab, Roboflow, Kaggle, Oracle APEX, VS Code, Git